

# UPMANYU JHA

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## WORK EXPERIENCE

### Quantiphi Analytics Solutions Private Limited (Google Cloud Partner)

Mumbai, IN

#### Machine Learning Engineer (LLMs, NLP, Computer Vision)

Jul 2023 - Present

**Scale & Architecture:** Refactored document Information Extraction and Retrieval into scalable microservices handling 80K+ documents quarterly, boosting classification accuracy by 19.5% and reducing manual review time by 80%

- Engineered a financial anomaly detection model for transaction integrity using statistical profiling, increasing recall by 35% and cutting manual review queues by 50% while maintaining high precision

**GenAI & Fine-Tuning:** Developed prompt engineering and fine-tuning frameworks for domain-specific LLMs, slashing model hallucinations by 85% via domain-specific fine-tuning and reduced GCP costs by 15% through advanced model compression techniques

- Achieved 92% F1-score in structured data extraction using few-shot prompting, reducing manual processing time by over 40%
- Designed a Prototype Retrieval-Augmented Generation (RAG) pipeline utilizing data vectorization techniques, fine-tuned LLMs and GCP ETL services, elevating financial document extraction F1-score from 0.55 to 0.90 (+63% improvement) to reduce manual auditing time

**Leadership & Delivery:** Led a cross-functional squad of 6+ engineers to brainstorm, design across 5+ business units with reusable custom artifacts; cut ramp-up time by 20% and improved team delivery scores by 15%

- Delivered 15+ GenAI pipelines on Google Cloud to production within 12-months technical roadmap, enforcing strict governance and software engineering principles around testing and code reviews, resolving scalability and reusability bottlenecks

**Model Serving & Monitoring:** Engineered a financial anomaly detection model using statistical data analysis and predictive modeling, cutting manual review queues by 50%; implemented continuous model monitoring and drift analysis frameworks to ensure production reliability

### Bugcrowd | Hackerone

Mumbai, IN

#### Security Researcher & Automation Engineer (Web Application, API)

Jun 2021 - Jul 2023

- System Engineering:** Developed robust Python, Bash and Agentic AI automation tools/workflows and infrastructure to scale application testing, boosting system evaluation efficiency by 60%
- Log Management:** Conducted Root Cause Analysis (RCA) on 30+ architectural vulnerabilities, gaining deep experience in handling performance, application, and security log management
- Software Reliability:** Scripted automated compliance checks and integrated them into deployment pipelines, ensuring high-fidelity code reviews and robust software engineering practices

## TECHNICAL SKILLS

- Programming & Tools:** Python, Go, SQL (PostgreSQL), C++, Git, Docker, Kubernetes, Linux, Postman
- Machine Learning & AI:** PyTorch, TensorFlow, LLMs (GPT, BERT), Prompt Engineering, Fine-tuning, NLP, Computer Vision, Scikit-Learn, Streamlit
- Generative AI & LLMs:** OpenAI, Gemini, Claude, **Fine-tuning** (LoRA), **Agents** (n8n, LangChain) **Hugging Face** (Transformers: BERT, LSTM)
- Cloud, MLOps & Infrastructure:** GCP (Vertex AI, GKE, BigQuery), AWS (Bedrock, Textract, SageMaker, S3), MLflow, VectorDBs (Pinecone, FAISS)

## PROJECTS AND RESEARCH

### Kisan-Setu - Agentic FPO (AWS AI for Bharat Hackathon) | AWS (Bedrock, Textract, SageMaker)

Feb 2026 - Mar 2026

- Architected an **offline-first serverless backend** with DynamoDB and AppSync, optimizing cloud costs to under \$50/month per FPO, ensuring reliable access and cost savings for rural organizations
- Engineered a **multimodal WhatsApp AI agent** utilizing AWS Bedrock and Amazon Transcribe to digitize rural agricultural operations, designed to support clusters of 500+ farmers by automating alternative credit scoring
- Implemented feature engineering and data augmentation workflows using Amazon Textract to extract financial entities from noisy, handwritten vernacular receipts, replacing manual data entry with a **'Zero-UI' system**
- Reduced expensive long-context AI prompting by 40% by implementing Bedrock Knowledge Bases, and cut SageMaker compute costs by up to 90% using Spot Instances for yield prediction models

### AstraShield - Disaster Relief Verification System | Python, FastAPI, Gemini Vision AI, React, GCP

Oct 2025 - Nov 2025

- Built a scalable **Q&A and Verification system** for safe donation verification using FastAPI and Gemini Vision, implementing Model Context Protocol (MCP) for Agentic function calling
- Engineered Forensic Agentic AI achieving a 90% detection rate via Human-in-the-Loop workflow across 100+ real-world samples, utilizing Gemini Vision Pro for multimodal analysis, delivering background checks and real-time risk scores in under 30 sec

## EDUCATION

### University Of Mumbai

Mumbai, IN

Bachelor of Engineering in Computer Engineering, **CGPA: 8.92/10** | **Relevant Coursework:** Deep Learning, Machine Learning, NLP, Cloud Computing

## CERTIFICATIONS AND TRAINING

Google Generative AI Leader (**GCP Gen AI**) | Google Cloud

Dec 2025

Google Cloud Professional Cloud Architect (**GCP PCA**) | Google Cloud

Jun 2025

Google Cloud Professional Machine Learning Engineer (**GCP PMLE**) | Google Cloud

May 2024